News Topic Classification

Meiling Wang **Ming-Hsiang Lee** **Ruoxuan Chen**
Tzu-Chieh Yang **Xinyan Huang**
College of Engineering
Northeastern University
Group 2

Abstract

Automated news classification is fundamental to search and recommendation systems, yet the conditions under which contextual language models outperform traditional keyword-based methods remain underexplored. This paper investigates when BERT fine-tuning provides meaningful gains over TF-IDF with Logistic Regression across two datasets with distinct structural characteristics: BBC News (1,194 articles, 5 categories, full-length text) and AG News (127,600 articles, 4 categories, short headlines). Our experiments show that BERT achieves 99.44% accuracy on BBC News and 93.87% on AG News, compared to the TF-IDF baseline at 97.78% and 91.78%, respectively. The improvement gap is larger on AG News (a 2.09 percentage-point increase) than on BBC News (1.66 percentage points), and detailed error analysis suggests that most remaining errors stem from inherent category overlap—particularly between Business and Sci/Tech—rather than model limitations. These findings suggest that dataset structure and category separability may play a more critical role than model architecture in determining performance for news classification tasks.

1 Introduction

Automated news classification supports a wide range of downstream applications, including content recommendation, search indexing, and news aggregation. Two major paradigms dominate this task: traditional bag-of-words methods such as TF-IDF combined with linear classifiers, and modern contextual language models based on transformers such as BERT (Devlin et al., 2019). While BERT consistently outperforms TF-IDF in general benchmarks, the conditions under which these improvements are practically meaningful remain underexplored.

Most prior work evaluates models on a single fixed dataset, making it difficult to disentangle the contribution of the model from the inherent difficulty of the task. Datasets with well-separated categories and long documents may favor both approaches, potentially masking the contextual advantages of BERT. In contrast, datasets with shorter texts and overlapping category semantics may better reveal where contextual understanding is essential.

This paper addresses the following central question: when does contextual language modeling provide a meaningful advantage over keyword-based classification? To investigate this, we apply both models using identical preprocessing and evaluation protocols to two datasets that differ along multiple dimensions—document length, category granularity, label semantics, and total size. We do not claim these datasets form a controlled comparison; rather, we treat their structural differences as a lens for interpreting where and why the performance gap between models changes. Our analysis goes beyond accuracy comparisons to examine per-class performance, confusion patterns, and specific misclassification examples.

Our main contributions are as follows:

- A cross-dataset comparison of TF-IDF+LR against fine-tuned BERT applied under consistent preprocessing and evaluation protocols.
- Empirical observations suggesting that dataset characteristics—particularly category overlap and text length—may contribute more to the observed performance gap than model architecture alone.
- An error analysis that surfaces recurring misclassification patterns, pointing toward label ambiguity as a potential contributing factor alongside model limitations.

2 Related Work

2.1 Traditional Text Classification

TF-IDF representations combined with linear classifiers such as Logistic Regression and Support Vector Machines have long served as strong baselines for text classification (Joachims, 1998; Yang and Liu, 1999). These methods represent documents as sparse vectors of token frequencies weighted by inverse document frequency, effectively capturing keyword salience when categories have distinctive vocabularies.

Despite their simplicity, TF-IDF-based models remain competitive in settings where training data is sufficient and category boundaries are well-separated. However, their reliance on surface-level features limits their ability to capture semantic meaning and contextual relationships between words.

2.2 Contextual Language Models

The introduction of BERT (Devlin et al., 2019) and subsequent transformer-based models enabled contextualized word representations that capture polysemy, syntax, and long-range dependencies. Fine-tuned BERT models have achieved state-of-the-art performance across a wide range of NLP tasks.

For text classification, prior work has shown that fine-tuning pre-trained language models significantly outperforms feature-based approaches (Sun et al., 2019; Howard and Ruder, 2018). However, these improvements are typically demonstrated on individual benchmark datasets, making it difficult to understand under what conditions contextual modeling provides the greatest benefit.

2.3 News Classification and Dataset Effects

News classification is a widely studied task due to its applications in search, recommendation, and information organization. The AG News dataset, introduced by (Zhang et al., 2015), has become a standard benchmark in this domain.

While prior work has primarily focused on comparing model architectures, less attention has been paid to how dataset characteristics—such as text length, category overlap, and label clarity—affect model performance. As a result, it remains unclear to what extent observed performance differences can be attributed to model design versus inherent dataset properties.

In contrast, our work directly investigates how dataset structure influences the relative performance of TF-IDF and BERT by evaluating both models under consistent preprocessing and evaluation protocols across two datasets with systematically different properties.

3 Datasets and Preprocessing

3.1 Datasets

We evaluate our methods on two datasets with contrasting structural properties.

BBC News (University College Dublin, 2004) is a widely used news classification dataset covering five categories: Business, Entertainment, Politics, Sport, and Tech. After preprocessing and filtering, 1,194 samples are used in our experiments. Articles are full-length, averaging approximately 380 words. We use a custom 70/15/15 train/validation/test split, yielding 835 training, 179 validation, and 180 test samples.

AG News (Zhang et al., 2015) is a large-scale benchmark dataset containing 127,600 news article titles and descriptions across four categories: Business, Sci/Tech, Sports, and World. Each sample averages approximately 8 words for the title and 50 words overall. We use the official test split of 7,600 samples and apply an 85/15 split to the training portion, yielding 102,000 training and 18,000 validation samples.

Property	BBC News	AG News
Size (articles)	1,194	127,600
Categories	5	4
Avg. length	∼380 words	∼50 words
Train split	70% (835)	85% (102,000)
Val split	15% (179)	15% (18,000)
Test split	15% (180)	Official (7,600)

Table 1: Dataset characteristics comparison.

3.2 Preprocessing Pipeline

We implement a unified preprocessing pipeline applied to both datasets. The pipeline consists of six stages:

1. **Dataset loading:** automatic detection of CSV or folder-based layout.

2. **Text cleaning:** removal of non-ASCII characters and leading/trailing whitespace. Original casing is preserved, although the BERT model used in this work (bert-base-uncased) does not rely on case information.

3. **Quality filtering:** removal of null values and duplicate entries, addition of a numeric `label_id` column for model compatibility, and generation of a lowercased `text_lower` column for the TF-IDF baseline.

4. **Data splitting:** BBC News uses a custom 70/15/15 stratified split; AG News uses the official test set and an 85/15 train/validation split.

5. **Debug subset:** extraction of 10 samples per class for rapid smoke testing.

6. **Output:** generation of four CSV files (train, validation, test, debug) committed to the shared repository.

4 Models

4.1 Baseline: TF-IDF + Logistic Regression

The baseline model converts lowercased article text into TF-IDF feature vectors using unigram and bigram features with sublinear term-frequency scaling. The resulting features are fed into a multinomial Logistic Regression classifier with L2 regularization. This approach relies entirely on surface-level keyword frequency patterns and does not capture word order or contextual information. Detailed hyperparameter settings are provided in Table 2.

Hyperparameter	Value
N-grams	Unigram + Bigram
Sublinear TF	True
Vocabulary size	50,000
Min document frequency	2
Max document frequency	0.95
Solver	lbfgs
Regularization (C)	1.0
Max iterations	2,000

Table 2: TF-IDF + Logistic Regression hyperparameters.

4.2 BERT Fine-Tuning

We fine-tune `bert-base-uncased` using the Hugging Face `BertForSequenceClassification` implementation, which applies a linear classification head over the `[CLS]` token. Inputs are tokenized with `BertTokenizerFast` and truncated or padded to 256 tokens for BBC News and 128 tokens for AG News.

The model is trained for 3 epochs using AdamW with a learning rate of 2×10^{-5} and weight decay of 0.01. A linear learning rate schedule with warmup is applied, and the checkpoint with the lowest validation loss is selected. Additional hyperparameters are provided in Table 3.

Hyperparameter	BBC News	AG News
Max sequence length	256	128
Epochs	3	3
Batch size	8	8
Learning rate	2×10^{-5}	2×10^{-5}
Weight decay	0.01	0.01
Warmup ratio	0.1	0.1
Optimizer	AdamW	AdamW

Table 3: BERT fine-tuning hyperparameters.

5 Experiments and Results

5.1 Overall Performance

Table 4 summarizes overall model performance across both datasets. BERT outperforms the TF-IDF baseline on both datasets, but the magnitude of improvement differs. On BBC News, BERT improves accuracy by 1.66 percentage points (97.78% $\rightarrow$ 99.44%). On AG News, the improvement is 2.09 percentage points (91.78% $\rightarrow$ 93.87%). The pattern holds for macro-averaged F1, with gains of +0.017 and +0.022 on BBC and AG News, respectively.

Notably, the same BERT model achieves 99.44% on BBC News but only 93.87% on AG News—a 5.57 percentage-point gap attributable to dataset characteristics, since the architecture and hyperparameters are held constant. This suggests that dataset difficulty may be a stronger determinant of absolute performance than model choice in this setting.

5.2 Per-Class Analysis on AG News

Table 5 reports per-class F1 scores for both models on AG News, where category overlap is most pronounced.

BERT provides the largest improvement for World (+2.6%) and the smallest for Sports (+1.6%). This pattern aligns with our hypothesis: Sports articles contain highly distinctive vocabulary (e.g., team names, sport-specific terminology, and scores), enabling TF-IDF to capture the signal effectively. In contrast, World, Business, and Sci/Tech articles share vocabulary across categories, where BERT's contextual encoding provides a meaningful advantage.

Dataset	Model	Acc.	Macro F1	W-F1
BBC News	TF-IDF + LR	97.78%	0.977	0.978
BBC News	BERT	**99.44%**	**0.994**	**0.994**
AG News	TF-IDF + LR	91.78%	0.918	0.918
AG News	BERT	**93.87%**	**0.939**	**0.939**

Table 4: Overall performance comparison across datasets.

Category	TF-IDF F1	BERT F1	Δ Gain
Business	0.887	0.908	+2.1%
Sci/Tech	0.895	0.916	+2.1%
Sports	0.967	0.983	+1.6%
World	0.921	**0.947**	**+2.6%**

Table 5: Per-class F1 scores on AG News.

5.3 Confusion Analysis

On BBC News, both models achieve near-perfect performance, as shown in Figure 1. The TF-IDF baseline makes only 4 errors, while BERT makes 1 error in the Entertainment category. This result indicates that BBC categories are highly separable using surface-level lexical features.

In contrast, Figure 2 shows the confusion matrices on AG News, where classification is more challenging. The dominant error pair for both models is Business $\leftrightarrow$ Sci/Tech. The TF-IDF baseline produces 153 Business $\rightarrow$ Sci/Tech errors and 118 Sci/Tech $\rightarrow$ Business errors, while BERT reduces these to 121 and 108, respectively. Although reduced, this confusion remains the most prominent, suggesting inherent category ambiguity.

6 Error Analysis

6.1 Category Overlap in AG News

To better understand the nature of model errors, we examined cases where both models make the same mistake. While our analysis is not exhaustive, several recurring patterns emerged, suggesting systematic tendencies rather than isolated failures. We highlight two illustrative examples below.

Example 1 (True: Sci/Tech, Predicted: Business). *"IBM to hire even more new workers. By the end of the year, the computing giant plans to have its biggest headcount since 1991."* Despite IBM being a technology company, the headline's dominant signals—hiring and headcount growth—are strongly associated with business discourse. Neither TF-IDF nor BERT can override this surface-level signal.

Example 2 (True: Business, Predicted: Sci/Tech). *"Broadband charges set to tumble. The cost of broadband internet access is likely to fall after Ofcom ordered British Telecom to cut the amount it charges internet providers."* The terms *broadband*, *internet*, and *telecom* appear technology-related, but the underlying topic—regulatory pricing decisions—is fundamentally a business issue. With only $\sim$50 words of context, BERT cannot reliably infer the regulatory framing.

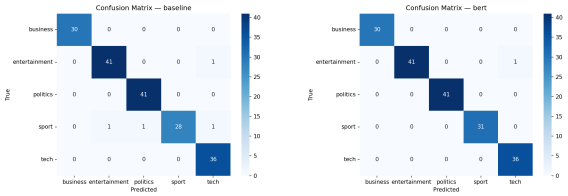

Figure 1: Confusion matrices on BBC News test set. Left: TF-IDF + Logistic Regression baseline. Right: fine-tuned BERT.

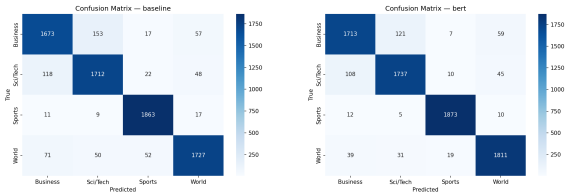

Figure 2: Confusion matrices on AG News test set. Left: TF-IDF + Logistic Regression baseline. Right: fine-tuned BERT.

These examples point to a recurring pattern: articles at the intersection of technology companies and business events are frequently misclassified by both models. This suggests that the Business/Sci/Tech boundary may be inherently ambiguous in short-form news, where a single headline often contains signals from multiple categories. We note that additional error patterns likely exist beyond those captured here, and a more comprehensive error audit could reveal further systematic tendencies.

6.2 Label Noise in BBC News

On BBC News, the single shared error between both models involves an article labeled *Entertainment* that discusses the Crossrail infrastructure project—a cross-London rail link connecting Heathrow Airport to Canary Wharf. The article's content (public funding, transport infrastructure, and journey time reduction) aligns more naturally with Tech or Politics than with Entertainment. This suggests a labeling inconsistency in the dataset, where the model prediction may better reflect the

true topic than the assigned label.

This example highlights the possibility that label quality contributes to some portion of classification errors. However, we cannot conclusively determine from a single case whether this reflects a systematic labeling issue or an isolated anomaly. Nevertheless, it illustrates that error analysis can surface potential annotation problems that are beyond the scope of model improvements, and that not all misclassifications are attributable to model limitations.

7 Discussion

7.1 When Does BERT Help?

Our results support a nuanced answer to the central research question. BERT's contextual advantage is most pronounced under three conditions: (1) categories share vocabulary and require understanding of semantic roles rather than keyword presence; (2) texts are short, making each token's contribution more critical; and (3) the classification boundary depends on discourse framing rather than topic keywords. These conditions are all present in AG News but largely absent in BBC News.

For BBC News, category-distinctive vocabulary is sufficiently strong that TF-IDF already achieves 97.78% accuracy. The marginal gain from BERT is therefore limited. In practical settings, the computational cost of BERT fine-tuning—orders of magnitude higher than TF-IDF—may not justify a 1.66 percentage-point improvement on similarly structured datasets.

7.2 Dataset Difficulty vs. Model Architecture

Perhaps the most striking finding is the 5.57 percentage-point accuracy gap between BERT on BBC News (99.44%) and BERT on AG News (93.87%), despite identical architecture and hyperparameters. Dataset characteristics appear to play a substantial role in determining classification performance, and may in some cases outweigh the effect of model sophistication.

We note, however, that because the two datasets differ along multiple dimensions simultaneously, these factors cannot be cleanly disentangled. Researchers comparing model performance across datasets should therefore account for this confound before attributing performance differences solely to architectural choices.

7.3 Systematic vs. Random Errors

A notable pattern is the extent to which TF-IDF and BERT share the same errors, particularly for the Business/Sci/Tech pair. This co-occurrence suggests that part of the residual error may stem from category ambiguity in the labeling schema rather than from model capacity alone.

We treat this as a preliminary observation. A more comprehensive analysis distinguishing shared and model-specific errors could provide deeper insight into how much of the error floor is inherently irreducible. Potential directions include hierarchical labeling schemes or multi-label annotation to better capture articles that span category boundaries.

8 Conclusion

We presented a systematic comparison of TF-IDF + Logistic Regression and fine-tuned BERT for news topic classification across two structurally different datasets. Our experiments show that BERT provides larger gains on short, semantically overlapping texts (AG News, +2.09%) than on longer, category-distinctive datasets (BBC News, +1.66%).

Our analysis further suggests that many remaining errors arise from inherent category ambiguity rather than model limitations. In particular, the same confusion patterns appear across both models in the most challenging cases, indicating that part of the error may be irreducible under the current labeling scheme.

Three practical conclusions follow from our findings. First, dataset difficulty—driven by category overlap, text length, and label quality—appears to play a more significant role than model architecture in determining absolute performance in our experiments. Second, BERT is most beneficial in settings where contextual understanding and semantic framing are required, rather than simple keyword matching. Third, persistent confusion between categories may indicate the need for improved labeling schemes rather than more complex models.

We note that these observations are based on two datasets and may not generalize to all classification settings. Future work could explore domain-adaptive pre-training, multi-label classification for boundary-spanning articles, and the effect of data augmentation on semantically ambiguous categories.

References

Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*.

Jeremy Howard and Sebastian Ruder. 2018. Universal language model fine-tuning for text classification. In *Proceedings of ACL*.

Thorsten Joachims. 1998. Text categorization with support vector machines. In *Proceedings of ECML*.

Chi Sun, Xipeng Qiu, Yige Xu, and Xuanjing Huang. 2019. How to fine-tune bert for text classification? ArXiv preprint arXiv:1905.05583.

University College Dublin. 2004. BBC News dataset. Accessed: 2026.

Yiming Yang and Xin Liu. 1999. A re-examination of text categorization methods. In *Proceedings of SIGIR*.

Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In *Proceedings of NeurIPS*.